Samson Koelle

Seattle, WA — sjkoelle@gmail.com — `sjkoelle.github.io`

I am a researcher and engineer working on artificial intelligence, interpretable machine learning, and computational biology. I take pride in programming and being a good teammate. I have published several first-author papers, and products I have worked on have been used by millions of people.

Skills: Generative AI, AI Agents, Machine Learning, Python, Computational Biology

Experience:

Amazon Web Services Applied Scientist II Seattle, WA 08/2024 – Present
R&D for AI agents for call centers

Uberduck, Inc. Cofounder / Head of ML Seattle, WA 08/2021 – 03/2024
Led AI R&D for text-to-speech start up

Allen Institute for Brain Science Graduate Assistant Seattle, WA 01/2020 – 12/2020
Data science and modeling for mesoscale connectomics

Amazon Research Intern Seattle, WA 06/2017 – 09/2017
R&D for supply chain forecasting models

Fred Hutchinson Graduate Assistant Seattle, WA 09/2015 – 06/2017
Cancer transcriptomics data analysis

National Heart, Lung, and Blood Institute IRTA Bethesda, MD 07/2013 – 05/2015
Experimental and computational hematopoiesis research

Education:

Y Combinator 01/2022 – 03/2022

Ph.D. Statistics University of Washington 09/2015 – 03/2022
Thesis: *Geometric Algorithms for Interpretable Manifold Learning*
Advisor: Marina Meila

B.A. Mathematics Columbia University 09/2009 – 05/2013

Recent publications:

Zhou, Yuqing, Zhuoer Wang, Jie Yuan, Hong Wang, Samson J Koelle, et al. (2026). "WISE-Flow: Workflow-Induced Structured Experience for Self-Evolving Conversational Service Agents". In: *arXiv preprint arXiv:2601.08158*.

Koelle, Samson J and Marina Meila (2024). "Isometry pursuit". In: *arXiv preprint arXiv:2411.18502*.

Koelle, Samson J, Hanyu Zhang, Octavian-Vlad Murad, and Marina Meila (2024). "Consistency of dictionary-based manifold learning". In: *Proceedings of the International Conference on Artificial Intelligence and Statistics*, pp. 4348–4356.

Koelle, Samson J, Dana Mastrovito, Jennifer D Whitesell, et al. (2023). "Modeling the cell-type-specific mesoscale murine connectome". In: *Network Neuroscience* 7.4, pp. 1497–1512.

Zhang, Hanyu, Samson J Koelle, and Marina Meila (2023). "Dictionary-based manifold learning". In: *arXiv preprint arXiv:2302.00263*.

Koelle, Samson J (2022). "Geometric Algorithms for Interpretable Manifold Learning". PhD thesis. University of Washington.

Koelle, Samson J, Hanyu Zhang, Marina Meila, and Yu-Chia Chen (2022). "Manifold coordinates with physical meaning". In: *Journal of Machine Learning Research* 23.133, pp. 1–57.

Xu, Jason, Samson J Koelle, Peter Guttorp, Chuanfeng Wu, Cynthia E Dunbar, Janis L Abkowitz, and Vladimir N Minin (2019). "Statistical inference for partially observed branching processes with application to cell lineage tracking of in vivo hematopoiesis". In: *Annals of Applied Statistics* 13.4, pp. 2091–2119.

Meila, Marina, Samson J Koelle, and Hanyu Zhang (2018). "A regression approach for explaining manifold embedding coordinates". In: *Stat* 7.1.

Paulson, Kelly G, Valentin Voillet, Michael S McAfee, Samson J Koelle, et al. (2018). "Acquired cancer resistance to combination immunotherapy from transcriptional loss of class I HLA". In: *Nature Communications* 9, p. 3868.

Wu, Chuanfeng, Diego A Espinoza, Samson J Koelle, et al. (2018). "Clonal expansion and compartmentalized maintenance of rhesus macaque NK cell subsets". In: *Science Immunology* 3.29, eaat9781.

Yabe, Idalia M, Lauren L Truitt, Diego A Espinoza, Chuanfeng Wu, Samson J Koelle, et al. (2018). "Barcoding of macaque hematopoietic stem and progenitor cells". In: *Molecular Therapy Methods & Clinical Development* 11, pp. 143–154.

Yu, Kyung-Rok, Diego A Espinoza, Chuanfeng Wu, Samson J Koelle, et al. (2018). "The impact of aging on primate hematopoiesis as interrogated by clonal tracking". In: *Blood* 131.11, pp. 1195–1205.

Koelle, Samson J, Diego A Espinoza, Chuanfeng Wu, et al. (2017). "Quantitative stability of hematopoietic stem and progenitor cell clonal output in rhesus macaques". In: *Blood* 129.11, pp. 1448–1457.

Wu, Chuanfeng, Diego A Espinoza, Samson J Koelle, Di Yang, Heinrich Schlums, Jan K Davidson-Moncada, Rong Lu, Amitinder Kaur, Hammer Quirin, Brian Li, et al. (2016). "Clonal Expansion and Long-Term Persistence of Rhesus Macaque NK Cells with an Adaptive Phenotype".

Xu, Jason, Samson J Koelle, Peter Guttorp, Chuanfeng Wu, Cynthia E Dunbar, Janis L Abkowitz, and Vladimir Minin (2016). "Stochastic modeling of hematopoietic stem and progenitor cell barcoding data from rhesus macaques challenges the classic model of hematopoiesis". In: *Blood* 128.22, p. 2643.

Wu, Chuanfeng, Samson J Koelle, Brian Li, Diego Espinoza, Rong Lu, Allen E Krouse, Mark Metzger, Robert E Donahue, and Cynthia E Dunbar (2015). "Clonal tracking of the geographic distribution of hematopoiesis in nonhuman primates". In: *Blood* 126.23, p. 241.

Jung, Moonjung, Jan K Davidson-Moncada, Chuanfeng Wu, Samson J Koelle, et al. (2014). "Specific impaired CD56+ NK cell development in GATA2 deficiency". In: *Blood* 124.21, p. 1419.

Koelle, Samson J, Chuanfeng Wu, Brian Li, Rong Lu, Robert E Donahue, and Cynthia E Dunbar (2014). "Long-term hematopoietic clonal stability tracked using molecular barcoding in non-human primates". In: *Blood* 124.21, p. 2908.

Wu, Chuanfeng, Brian Li, Rong Lu, Samson J Koelle, Yanqin Yang, Alexander Jares, Alan E Krouse, Mark Metzger, Karin Loré, et al. (2014). "Clonal tracking of rhesus macaque hematopoiesis highlights a distinct lineage origin for natural killer cells". In: *Cell Stem Cell* 14.4, pp. 486–499.

Software

`https://github.com/uberduck-ai/uberduck-ml-dev`	Audio ML models	300+ stars
`https://github.com/sjkoelle/montlake`	Geometric data analysis	10+ stars